
UNIVERSITY OF THE PHILIPPINES DILIMAN
College of Science
National Institute of Physics

Physics 180 THV-1
Problem Set 1

VERCIL JUAN
IV - BS Physics

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Problem 1.1

Go to **arXiv** and have a look at the relevant subjects for this course (nucl-ex, nucl-th, hep-ex, hep-ph, hep-th).

1. What do the first four digits of the **arXiv** number unique to a specific paper stand for?
 2. What are the **arXiv** numbers for the papers shown in Slides 22 and 23.
 3. Cite a single paper that fascinates you within the dates **August 11-20, 2026**, and briefly explain why you find this particular paper interesting.
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Solution

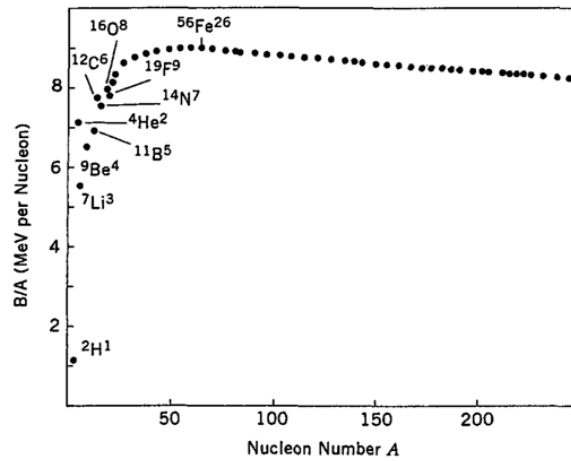
Problem 1.2

Go to InSpire and search for all papers from 1967. What is the most highly-cited paper from that year? Skim through its introduction. Why do you think is it highly cited?

Solution

Problem 2.1

The B/A plot looks like it peaks at a specific value of A . Using the curve approximated by SEMF, find the value of A that achieves this and find the peak value. What is the element that achieves this



Solution

References

Appendices

Code

No code was used for this problem set